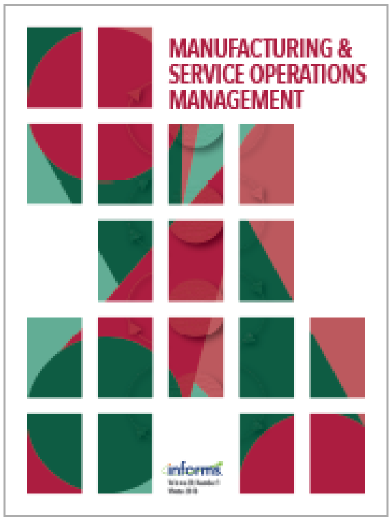

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An Empirical Investigation of Ridesharing and New Vehicle Purchase

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Abstract. ***Problem definition*:** In this work, we examine how ridesharing platforms affect changes in short-term vehicle purchasing. On the one hand, if the introduction of such platforms motivates owners to use the idle capacity of their existing vehicles to accrue rents, vehicle sales might fall. On the other hand, if the platform induces would-be drivers to purchase new vehicles in order to participate, vehicle sales might rise. ***Academic/practical relevance*:** Whereas operations management researchers have begun to broach this subject analytically, this work provides empirical evidence of the impact of ridesharing platforms on new vehicle ownership. Further, we assess heterogeneity in the effect across vehicle type and location. ***Methodology*:** We examine this tension using a unique data set of new vehicle registrations in China. In doing so, we exploit the variation in the timing of Uber entry using a difference-in-difference approach. ***Results*:** Findings suggest Uber entry is associated with a significant short-term increase in private new vehicle ownership, indicating that consumers actively change their stock of held resources to capture excess rents offered by the platform. These effects exclusively manifest among vehicle brands that qualify for the platform. Further, inasmuch as sales of vehicles with smaller displacement increase more than large-displacement vehicles, results indicate that the effect of Uber entry varies considerably across vehicle types. Finally, results indicate that the effects are stronger in locations where established public transportation options are weaker. ***Managerial implications*:** Results provide initial evidence that manufacturers can benefit from the emergence of the sharing economy, especially manufacturers whose products align with the needs of platform participants. For policy makers, our findings further undercut claims made by platforms that the individuals working on them are exploiting already existing resources, suggesting some form of nascent professionalism on the part of platform workers.

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Keywords: sharing economy • platform • ridesharing • car sales • durable goods • difference in difference • relative time model

1. Introduction

The rise of the sharing economy has dramatically changed the way consumers and service providers interact (Sundararajan 2016). Still, researchers are scrambling to determine how platform-enabled businesses affect behavior at the societal, firm, and personal levels (Greenwood et al. 2017). A host of recent research attests to such changes, including impacts on incumbent industries (Wallsten 2015, Zervas et al. 2017, Chu et al. 2021), increased efficiency (Cramer and Krueger 2016), improved public safety (Greenwood and Wattal 2017), and changing access to job opportunities (Burtch et al. 2018). Yet, within this expanding corpus of work, scholars primarily focus on the demand side of the economy, that is, those who utilize such services instead of those who provide such services, with some notable exceptions, such as the scholarship on temporal microdynamics and incentives, such as surge pricing (e.g., Chen and Sheldon 2015, Kabra et al. 2016, Cachon et al. 2017, Allon et al. 2018, Barron et al. 2018, Burtch et al. 2018, Karacaoglu et al. 2018, Guda and Subramanian 2019). Even within these exceptions, scholars have yet to consider how consumers may be updating their behavior to capture the rents offered by this economy, which has grown significantly in recent years.

Moreover, whereas platforms argue that they are simply brokerages for individuals to share the idle capacity of their assets, opponents argue that individuals who participate on such platforms are making capital investments to capture rents as professionals, thereby circumventing the regulatory restrictions that govern traditional firms (Malhotra and Van Alstyne 2014). Indeed, theoretical work finds that the introduction of a platform can lead to both positive and negative impacts on allocative efficiency (Lee and Whang 2002). In this study, we empirically open this question by asking does the entry of ridesharing platforms yield a material short-term change in the rate of durable goods (viz., automobile) purchase?

The answer to this question is far from clear. On the one hand, if the introduction of such platforms motivates owners to use the idle capacity of their existing vehicles to accrue rents, vehicle sales might fall. And, although prior work offers conflicting definitions of the term "sharing economy," there is relative scholarly consensus that, in order for these markets to function, there must be some degree of asset underutilization (Schlagwein et al. 2020). Indeed, sharing platforms across the globe argue that supply-side participants are simply unlocking the value of at-rest resources in order to recoup rents from inefficiently used assets (Bhuiyan 2016, Bull 2018). Consequently, the sharing economy may lead to increased access and higher utilization of assets, thereby improving consumer welfare and reducing societal costs (e.g., pollution) (Benjaafar and Hu 2020).

On the other hand, if these platforms induce would-be drivers to purchase new vehicles, vehicle sales might rise. Despite the commonality of the belief that the sharing economy involves sharing underutilized resources, numerous researchers raise the possibility that supply-side participants may be strategically investing in order to capture the excess rents provided by the platform, thereby leading to value enhancement and stimulating new ownership (Jiang and Tian 2016, Abhishek et al. 2021). There are at least three reasons we might expect such an effect. First, would-be drivers may purchase vehicles to capture additional income by participating in the sharing economy inasmuch as such platforms imbue owners with the capacity to earn additional income (Cachon et al. 2017, Burtch et al. 2018, Hall and Krueger 2018). Second, would-be drivers may elect to upgrade their current vehicles and defray the cost of the new vehicle through the platform. Third, would-be drivers may be compelled to replace outdated vehicles that do not meet platform standards to participate. Therefore, besides joining the sharing platform by using idle assets, the platform may stimulate short-term purchasing if these reasons impact vehicle ownership decisions.

Complicating any intuition further is that participants on the demand side may also change their stocks of resources (Fraiberger and Sundararajan 2015). Consumers who use these platforms may idle their vehicles, postpone a planned vehicle purchase, or even jettison vehicles entirely as alternative transportation becomes available. As these platforms offer low-cost transportation (Cramer and Krueger 2016), consumers may reduce their personal vehicle utilization by leveraging the mobility the platform offers. Thus, for consumers on the margin between replacing and retaining a vehicle, it is plausible that such a purchase is delayed. And, although transportation choices are difficult to change (Oakil et al. 2011), it is possible that marginal consumers may postpone their purchase decisions when a ridesharing service becomes available.

Taken in sum, the effect ridesharing platforms have on vehicle ownership decisions is deeply murky. Whereas researchers have begun to broach this subject analytically (Fraiberger and Sundararajan 2015, Jiang and Tian 2016, Benjaafar et al. 2018, Filippas et al. 2020, Abhishek et al. 2021), limited empirical evidence exists of the impact of ridesharing platforms on new vehicle ownership.

To answer this question, we exploit the variation in the timing of Uber's entry into different cities in China. In August of 2014, Uber's peer-to-peer ridesharing service (People's Uber) was started. We estimate the effect of Uber entry on vehicle purchases using a unique data set of new vehicle registrations by private owners between 2010 and 2015. As Uber entry is temporally and geographically dispersed, we take a difference-in-difference approach. Results suggest that Uber entry is correlated with an immediate rise in new vehicle ownership, indicating that value enhancement dominates any idling by passengers in the short term.

To better identify the underlying mechanism behind this short-term spike, we explore heterogeneity in the effect based on vehicle type and location. Findings are threefold. First, the effect only manifests for vehicles that meet Uber's minimum price requirement with no material effect on vehicles below the price threshold. This suggests that the effect is primarily driven by the demand induced by Uber entry. Second, there is a stronger effect on lighter vehicles. This corroborates the value-enhancement mechanism because the platform incentivizes the purchase of vehicles with superior fuel economy (and, thus, larger rents). Finally, results indicate that the effect is stronger in locations with less developed transportation options. This, once again, corroborates the value-enhancement mechanism as consumers with fewer outside options are more likely to use an emerging transportation option (Babar and Burtch 2020).

Several contributions stem from this work. First, we contribute to the emerging stream of literature on the sharing economy by providing an initial examination of how the changing presence of ridesharing affects durable goods purchase (Jiang and Tian 2016, Benjaafar et al. 2018, Filippas et al. 2020, Abhishek et al. 2021). Whereas consumers might idle their vehicles to use these platforms, findings suggest that actors are sufficiently changing their stock of the product to capture emerging rents. Because of data availability, we are unable to estimate the long-term effect. However, prior analytical work shows that, in equilibrium, it is possible for ridesharing platforms to increase vehicle ownership in the long run (Benjaafar et al. 2018). In this same vein, our work extends research on the societal impacts of the sharing economy by considering how durable asset stocks are changing (Cohen et al. 2016, Greenwood and Wattal 2017, Burtch et al. 2018) and the impacts on incumbent industries (Wallsten 2015, Zervas et al. 2017, Cui et al. 2020b, Chu et al. 2021).

This study also has implications for manufacturers. Manufacturers may benefit from the presence of the sharing economy as these platforms result in an immediate and observable rise in sales. Coupled with finite budgets, our findings have implications for where those resources should be allocated. Results suggest resources should be allocated toward building smaller vehicles with superior fuel economy when platforms enter the market. To the extent that these vehicles are more often leveraged by ridesharing drivers, they are more likely to capture the glut in purchasing. Finally, findings reveal data on the regions to which newly manufactured vehicles should be allocated. Insofar as the effect is stronger in areas with weak public transit systems, results underscore this is where resources should be devoted. Thus, there are incentives for manufacturers to support the entry of sharing platforms and target consumers who are more likely to participate.

Finally, with respect to the current policy debate on the legality of sharing platforms (Malhotra and Van Alstyne 2014), findings suggest that the sharing economy may create externalities in the form of durable goods purchase. Coupled with evidence of other benefits of the sharing economy, for example, higher efficiency and productivity (Cramer and Krueger 2016) and higher consumer surplus (Cohen et al. 2016), the externality may be nontrivial. It is worth noting that our findings also undercut claims made by platforms that workers are simply exploiting at-rest resources. It is evident that platform workers are making capital investments to exploit these platforms (in the form of vehicle purchase), suggesting a nascent professionalism on the part of platform workers. This buoys claims made by critics, who allege that platform participants are engaging in regulatory arbitrage to bypass the red tape surrounding licensed livery and other regulated transportation services (Malhotra and Van Alstyne 2014).

2. Related Literature and Hypotheses

2.1. Sharing Economy

The sharing economy has received considerable attention from researchers since its emergence. Broadly, prior literature in this space can be grouped into three streams: platform design, impact on incumbent industries, and the societal impacts of these platforms. Each is described.

Focus on the design of sharing platforms is primarily on the means by which matching efficiency or consumer utility might be increased (Einav et al. 2016, Fradkin et al. 2017) and the impacts of platform design (Chen and Sheldon 2015, Cohen et al. 2016, Cachon et al. 2017, Taylor 2018, Bai et al. 2019, Cui et al. 2020a). A small number of studies, for example, analyze Uber's platform design, finding that the surge pricing utilized by Uber both significantly increases labor supply (Chen and Sheldon 2015) and generates substantial consumer surplus (Cohen et al. 2016, Cachon et al. 2017, Lam and Liu 2017).

When considering the impact on incumbent industries, results are equally rich (Wallsten 2015, Zervas et al. 2017, Babar and Burtch 2020, Cui et al. 2020a, Chu et al. 2021). Wallsten (2015), for example, examines the effect of Uber entry on the taxi industry, finding an increase in Uber popularity is associated with a decline in the number of complaints about taxis; underscoring the disruptive capability of platforms. Babar and Burtch (2020) find that ride-hailing services reduce the utilization of city bus services but increase the utilization of commuter rail by solving last-mile problems. Chu et al. (2021) and Cui et al. (2020b) study the impacts of bike sharing, which eases the last-mile problem, on subway housing premiums and restaurant pricing, respectively. Zervas et al. (2017) investigate the effect of Airbnb on the hoteling industry; finding an increase in Airbnb listings is associated with a decrease in hotel revenue. This research also continually shows that lower quality vendors are displaced by the entrance of sharing platforms, indicating a nontrivial price sensitivity of participants in these marketplaces (Greenwood and Wattal 2017, Zervas et al. 2017).

The third stream examines societal impacts of the sharing economy. Despite concerns about the negative impacts of the sharing economy (Malhotra and Van Alstyne 2014), empirical research expresses a balanced view. Greenwood and Wattal (2017), for example, find that Uber entry is associated with a decline in drunk-driving fatalities in California. Scholars also show that sharing platforms generate nontrivial consumer welfare (Cohen et al. 2016, Cramer and Krueger 2016). Negative outcomes are also observed, such as gender and racial discrimination (Edelman et al. 2017, Cui et al. 2020a). It is interesting that such bias is observed against both passengers (Mejia and Parker 2021) and drivers (Greenwood et al. 2022).

Yet, whereas prior research extensively documents the effects wrought on the demand side of the sharing economy, research considering the supply side is sparser. And, within the body of work that examines the supply side of the economy, focus is primarily on the characteristics of likely participants (Hall and Krueger 2018) or how temporal microdynamics and incentives (e.g., surge pricing) influence the decision to participate more or less (Chen and Sheldon 2015, Kabra et al. 2016, Cachon et al. 2017, Allon et al. 2018, Karacaoglu et al. 2018) rather than what steps a decision maker might take to access the sharing economy. This is problematic because it is not clear whether suppliers are using existing durable goods to participate on those platforms or making new investments to capture rents (Malhotra and Van Alstyne 2014).

2.2. Hypothesis Development

The impact of secondary markets on the value of assets has long been a source of scrutiny by scholars (Lee and Whang 2002). And, whereas several papers examine the impact of sharing platforms on purchase analytically (Jiang and Tian 2016, Benjaafar et al. 2018, Filippas et al. 2020, Abhishek et al. 2021), empirical study is limited. Unlike traditional markets, sharing platforms are unique because both the demand and supply sides are consumers of the durable good (Benjaafar and Hu 2020, Filippas et al. 2020). The supply side consists of owners who participate in the economy to capture rents (Burtch et al. 2018). The demand side consists of consumers who use the platform as an alternate means to "rent" the good (Filippas et al. 2020, Abhishek et al. 2021). We next discuss how supply- and demand-side effects may affect automobile sales.

2.2.1. Purchase Decline or Delay. How might the entry of ridesharing platforms, for example, Uber or Lyft, influence vehicle ownership decisions of supply-side participants? On the one hand, the core business model of the sharing economy is one in which underutilized resources are shared among peers (Schlagwein et al. 2020). If this is the case, ridesharing platforms are serviced by individuals who already own vehicles with excess capacity (Hall and Krueger 2018), meaning that vehicle purchase decisions will likely be unaffected by platform entry.

Received research further indicates that ridesharing platforms provide efficient and low-cost transportation options (Belk 2014, Cramer and Krueger 2016, Greenwood and Wattal 2017). Thus, inasmuch as riders exploit these new options, lowering or effectively limiting the utilization of their preexisting transportation options, it is intuitive that they substitute away from personal transportation (Babar and Burtch 2020). Should the transportation option from which they shift away be the use of their own vehicles, short-term vehicle sales may even fall. Coupled with the fact that suppliers may exploit excess capacity without accelerating their purchase timing, it is plausible that overall vehicle purchasing may decline.

Digging deeper, to the extent that vehicle ownership can be financially and emotionally burdensome (Schaefers et al. 2016), these platforms may limit a consumer's propensity to even own a car. Put simply, there are at least three reasons some consumers may no longer wish to subsidize the cost of vehicle ownership. First, there may be significant economic savings from exclusively leveraging ridesharing options. In addition to eliminating the costs of upkeep and insurance (Bardhi and Eckhardt 2012), ridesharing may be more attractive for consumers who do not want to curtail their social activities as a result of needing to drive, that is, drinking. Second, access-based consumption is often more convenient, flexible, and adaptable (Bardhi and Eckhardt 2012). Third, access-based consumption allows consumers to experiment with products and increase the diversity of goods consumed (Filippas et al. 2020), which may be perceived as a trendy or environmentally friendly (Belk 2014). Indeed, the popular press identifies several cases of people opting out of vehicle ownership as a result of ridesharing availability, and recent surveys find that people who use shared transportation modes are less likely to own a car and spend less on transportation (Manjoo 2014).

Finally, even though transportation choices tend to be difficult to change in the short term (Oakil et al. 2011), it is plausible that a vehicle replacement is delayed because of a changing perceived value of and reliance upon the vehicle for customers on the margin. The decision to replace a vehicle is often based on risk avoidance, notably for late replacers (Betts and Taran 2006). As a result, the idling of the vehicle in favor of ridesharing may lead consumers to prolong ownership because of a prolonged window of usability. Because of diminished reliance on the vehicle, it is likely that the perceived risk of events such as breakdowns also falls, thereby extending the purchase deliberation period for replacement or obviating it entirely (Putsis and Srinivasan 1994). Thus, from the demand perspective, ridesharing platforms may substitute for public transportation by diminishing reliance on personal automobiles (Babar and Burtch 2020). Therefore, we have the following.

Hypothesis 1a. *Entry of a ridesharing platform is negatively correlated with short-term vehicle sales.*

2.2.2. Stimulated Purchase. On the other hand, the presence of sharing platforms may lead to a value-enhancement effect, wherein a change in the market significantly increases the value of ownership (Jiang and Tian 2016, Abhishek et al. 2021). If this is the case, consumers may adjust their valuation of a vehicle by taking into account the rents that can be captured on these platforms, viz., the possibility of sharing excess capacity to earn additional income (Chen and Sheldon 2015, Cachon et al. 2017, Burtch et al. 2018, Hall and Krueger 2018). Such an effect may alter consumer incentives to purchase, inducing new ownership (Jiang and Tian 2016, Benjaafar et al. 2018, Filippas et al. 2020), because ridesharing platforms imbue owners with the possibility of sharing excess capacity and earning additional income (Chen and Sheldon 2015, Hall and Krueger 2018).

Empirical evidence lends support for these possibilities. Recent surveys find that 8% of Uber drivers in the United States were unemployed before they started working through Uber (Hall and Krueger 2018). For 20%, Uber is their only source of income. Still, most Uber drivers choose the platform as an additional source of income (91%), for a flexible work schedule (87%), and as a way to maintain stable income (74%). Burtch et al. (2018) find that the entry of Uber has a negative effect on entrepreneurship by providing stable employment for the unemployed and underemployed, thereby creating value. Thus, if value enhancement exists, participants may be willing to make capital investments in durable goods to access these markets.

Formally, value enhancement is likely to increase new vehicle purchases in multiple ways. First, would-be drivers who do not own a vehicle may be newly incentivized to purchase a vehicle if they wish to capture the rents offered by the platform. Second, would-be drivers may choose to upgrade their current vehicles by defraying the cost of the new vehicles through the platform. Third, vehicle owners may increase the utilization of their vehicles when driving for the platform, thereby requiring replacement by accelerating consumption and depreciation. Fourth, would-be drivers, if they are existing vehicle owners, may be compelled to replace or upgrade their current vehicles when the platform enters to meet the standards to participate on the platform. As is expected, Uber and other platforms maintain nontrivial vehicle standards for participants. In China, Uber requires vehicles to be less than five years old and valued at 80,000 RMB. And, whereas vehicle purchasing has been increasing in China over the last two decades, this represents a nontrivial portion of the market, notably among would-be drivers, thereby necessitating de novo purchasing. Finally, platforms often offer nontrivial incentives (e.g., loans and guaranteed compensation) as a way to seed the network, which directly manipulates vehicle valuation (e.g., the now defunct Uber financing program that offered loans to drivers to facilitate vehicle purchase). Therefore, we have the following.

Hypothesis 1b. *Entry of a ridesharing platform is positively correlated with short-term vehicle sales.*

It should be noted that these hypotheses are not mutually exclusive. On the supply side, it is unclear whether the platform mainly attracts existing owners utilizing their underutilized vehicles or incentivizing would-be drivers to invest in new vehicle ownership. On the demand side, it is unclear whether the presence of the platform leads consumers to delay or forgo their vehicle purchase decisions. Indeed, much of the analytical literature attests to the fact that these competing theoretical predictions may concomitantly manifest (e.g., Jiang and Tian 2016). In what follows, our goal is to determine which effect dominates.

3. Context and Data

3.1. Context

To estimate the effect of platform entry on vehicle purchase, we leverage the entry of Uber into cities in China. Founded in 2009 in San Francisco, Uber is currently the largest ridesharing service in the world by both valuation and footprint. Empirically, we focus on the Chinese counterpart of the discount Uber X service, People's Uber. Consistent with prior work (e.g., Burtch et al. 2018), we focus on discount services as opposed to premium services because of their larger networks of drivers and smaller barriers to entry. Until its purchase by DiDi in August of 2016, People's Uber was the largest peer-to-peer ridesharing platform in major cities in China. To use the service, riders request a trip through the app. Uber then assigns drivers who use their own vehicles to fulfill the request. An entry schedule of People's Uber can be found in Table A1, Online Appendix A.

We focus on People's Uber for several reasons. First, the introduction of People's Uber in China is geographically and temporally dispersed, allowing us to leverage a difference-in-difference design. Second, when Uber enters a new city, the company makes announcements on its blog and social media platforms. This allows us to cleanly identify the entry time for each city. It also differs from other ride-hailing platforms, such as DiDi, which do not usually make formal announcements when entering new regions. Third, Uber was the first platform to offer peer-to-peer ridesharing services in China, and received knowledge indicates that it was the market leader in peer-to-peer ridesharing services in China during the sample (Kalanick 2015, Mohan 2015). This separates it from other platforms that offered taxi-hailing services (viz., DiDi Chuxing) or black car services (e.g., DiDi Zhuanche, ShenZhouZhuanChe, and Yidao Yongche) during the sample.

Three complicating factors should be noted. First, among cities with People's Uber, three cities (Wuhan, Tianjin, and Nanjing) first launched the service unofficially, during which there was limited availability. We discuss our empirical strategy to resolve this concern subsequently. Second, seven cities imposed registration lotteries or auctions to limit the number of passenger cars, thereby driving down vehicle purchase. We exclude these cities. Finally, in May 2015, DiDi Chuxing introduced a competing peer-to-peer ridesharing service, DiDi Kuaiche. Whereas People's Uber remained the dominant discount service, this is empirically worth noting. Although the appearance of an additional ridesharing service does not undermine our theoretical arguments, because DiDi is subject to the same theoretical mechanisms as People's Uber, we do control for the potential effect of DiDi's ridesharing service (discussed subsequently).

3.2. Data

We construct a longitudinal data set of new car registrations in China from 2010 to 2015, which capture the monthly number of new passenger vehicles registered for personal use in each prefecture-level city. The source of these data are the transportation authority of a major city in China (subject to a nondisclosure agreement). Our data only include passenger cars manufactured in China. Empirically, this is of little concern. During the sample, imports constituted less than 3.1% of the Chinese passenger car market and mostly comprised high-end luxury vehicles (Li et al. 2015). The personal-use distinction is also critical as Uber exclusively allows individuals with their own vehicles to drive on the platform prior to 2016. Finally, these data do not include used cars. This is also of little concern as the Chinese used passenger car market was roughly one fifth the size of the new car market during the sample, and only a small portion of the used car market met Uber's criterion, that is, manufactured within five years (about 30% of used passenger cars were older) with a value no less than 80,000 RMB (about 75% of used passenger cars were priced lower) (Cai 2019, China Automobile Dealers Association 2019, Forward 2019).

3.3. Variable Definitions

3.3.1. Dependent Variable. The dependent variable, $ln(NumCars)_{it}$, is the natural log of the number of new vehicles registered in city i during month t plus one. The log transformation allows us to interpret the effect as an elastic percentage change and resolves the issue of a right-skewed distribution.

3.3.2. Independent Variables. People's Uber rolled out in two phases: the unofficial and then official launch. The unofficial launch did not occur in all cities. $Official\text{-}Uber\text{-}Entry_{it}$ is a dichotomous variable indicating the official entry of People's Uber in city i in month t. Using this treatment, we create a series of relative time indicators capturing the temporal distance between month t and Uber entry. This allows us to estimate a popular variant of the Autor (2003) leads and lags estimation, which permits us to probe the validity of the parallel trends assumption (discussed further in Section 4). We also consider $Unofficial\text{-}Uber\text{-}Entry_{it}$, which indicates unofficial entry of the service in city i in month t. $Unofficial\text{-}Uber\text{-}Entry_{it}$, if it occurs at all, always occurs before $Official\text{-}Uber\text{-}Entry_{it}$. The two variables are mutually exclusive.

3.3.3. Controls. In addition to these variables, we include a robust set of controls. First, we include annual city-level demographic data from the China City Statistical Yearbook, which is widely used in prior work (e.g., Jia 2014, Glaeser et al. 2017). Economic controls include population, the number of employed workers, the number of unemployed citizens, the average wage of workers, and gross region production (GRP) per capita. Further, because Uber is primarily app-based, we include the number of mobile and internet subscribers. Next, we control for transportation-related factors, including highway passenger traffic, the number of buses, bus passenger volume, the number of taxis, road area per capita, and metro length. Finally, we collect data on monthly gasoline price for each city from a portal website for professional investments (cngold.org) to account for the impact of gasoline prices on drivers' participation willingness.

To control for the influence of other ride-hailing services, we collect data from the Baidu index for each major competing ride-hailing service at the city–month level. This is done to capture the popularity of each platform in the local market. This is the Chinese equivalent of Google Trends data, a widely used proxy for demand (Wallsten 2015). Thus, *ln(Didi)* is the log of the Baidu index of search terms related to DiDi Chuxing (viz., DiDi Chuxing, Didi Dache, Kuaidi Dache, and Didi Kuaiche), *ln(Shenzhou)* is the log of the Baidu index of the search term ShenZhouZhuanChe, and *ln(Yidao)* is the log of the Baidu index of the search term YiDao YongChe. Note that Lyft, Sidecar, and other dominant U.S. competitors did not operate in China during the sample. Summary statistics are in Table 1, and a correlation matrix is in Online Appendix A.

4. Empirical Strategy

As discussed, to estimate the effect of Uber entry on short-term changes in durable goods purchase, we use a popular variant of the leads and lags model proposed by Autor (2003), which is widely used in extant literature in economics (e.g., Chung et al. 2017, Fetzer 2019), operations management (e.g., Pierce et al. 2015, Brynjolfsson et al. 2019), and studies of the effects of internet platforms (Greenwood and Wattal 2017, Burtch et al. 2018). The model is implemented by creating a series of dummies that indicate the temporal distance between an observation period, t, and Uber entry in city i in addition to the absolute year–month fixed effects. The relative time model offers two benefits. First, it allows us to assess any differences across treatment and control before treatment occurs. The absence of observed pretreatment differences bolsters the case that Uber entry can be treated as an exogenous event (once conditioned upon controls). Second, it allows us to estimate treatment effects over time (Wolfers 2006). Relative time indicators three or more quarters after treatment are collapsed into a single treatment indicator to avoid power issues. Cities where Uber has not yet entered serve as control for cities where Uber is active. The first sample comprises cities where Uber had entered by the end of 2015.

We begin the estimation with a base model absent controls or city-specific trends. We then include city-specific linear trends. Finally, we add the controls. We

Table 1. Summary Statistics

	Panel A: Treated cities[a]					Panel B: Untreated cities[b]				
Variable	Observations	Mean	Standard deviation	Minimum	Maximum	Observations	Mean	Standard deviation	Minimum	Maximum
Unofficial-Uber-Entry	1,008	0.009	0.094	0	1	18,853	0	0	0	0
Officia-lUber-Entry	1,008	0.097	0.296	0	1	18,853	0	0	0	0
ln(NumCars)	1,008	9.433	0.535	7.739	10.984	18,853	7.474	0.925	2.565	10.656
ln(Didi)	1,008	4.679	4.907	0	11.822	18,853	3.489	3.970	0	11.792
ln(Yidao)	1,008	5.003	3.562	0	10.035	18,853	2.619	3.085	0	10.415
ln(Shenzhou)	1,008	1.483	3.323	0	10.049	18,853	1.030	2.446	0	9.588
ln(Population), in 10,000	1,008	6.533	0.595	5.182	8.125	18,784	5.815	0.675	3.020	7.122
ln(Employed), in 10,000	1,008	5.029	0.505	4.034	6.862	18,772	3.440	0.645	1.710	5.505
ln(Registered Unemployed)	1,008	11.009	0.506	9.843	11.940	18,652	9.672	0.716	4.968	13.306
ln(Average Wage), in RMB	1,008	10.765	0.214	10.341	11.255	18,592	10.484	0.275	9.451	11.366
ln(GRP Per Capita), in RMB	1,008	11.209	0.396	10.04	12.201	18,760	10.339	0.610	4.605	12.579
ln(Mobile Subscribers), in 10,000	1,008	7.008	0.406	5.874	7.860	18,784	5.472	0.689	2.863	7.524
ln(Internet Subscribers), in 10,000	1,008	5.247	0.540	4.182	7.337	18,532	3.514	0.810	0.023	6.377
ln(Highway Passenger Traffic), in 10,000	1,008	10.101	0.831	8.110	12.016	18,700	8.608	0.904	0.519	12.566
ln(Buses)	1,008	8.523	0.416	7.414	9.400	18,652	6.228	0.948	3.135	13.292
ln(Bus Passenger Volume), in 10,000 person-times	1,008	11.387	0.522	10.180	12.397	18,568	8.784	1.189	3.466	12.437
ln(Taxis)	1,008	8.918	0.622	7.670	9.925	18,640	7.233	0.915	2.580	9.853
ln(Road Area Per Capita), in m^2	1,008	2.728	0.345	1.917	3.252	18,580	2.374	0.529	0.270	6.096
ln(Metro Length), in km	1,008	1.975	2.012	0	5.313	18,853	0.045	0.410	0	4.754
ln(Gasoline 90 Price), in RMB/L	1,008	1.890	0.090	1.278	2.111	18,827	1.889	0.094	1.273	2.128

Notes. The sample contains data from 2010 to 2015 (72 months). The observations are at the city–month level.
[a]For treated cities, we use 14 cities that experienced People's Uber entry and did not have car purchase restrictions.
[b]For untreated cities, we use 262 prefecture-level cities in our sample that (i) did not experience People's Uber entry and (ii) did not have car purchase restrictions.

use the following specification for the complete model with city-specific linear trends and controls:

$$\begin{aligned} ln(NumCars)_{it} = \alpha_i + \gamma_t + \sum_j \tau_j Pre\text{-}Uber\text{-}Entry_{it}(j) \\ + \delta Unofficial\text{-}Uber\text{-}Entry_{it} \\ + \varphi Official\text{-}Uber\text{-}Month_{it} \\ + \sum_k \omega_k Post\text{-}Uber\text{-}Entry_{it}(k) \\ + \boldsymbol{X}_{it}\boldsymbol{\beta}_2 + \theta_i t + \varepsilon_{it}, \end{aligned} \quad (1)$$

where α_i is a city fixed effect and γ_t is a calendar fixed effect at the year-month level. The indicator $Pre\text{-}Uber\text{-}Entry_{it}(j)$ equals one if the temporal distance between Uber's initial entry into city *i* and month *t* is *j* (in three-month intervals to increase interpretability and statistical power (Burtch et al. 2018); results are consistent using month indicators (Online Appendix C)). The indicator $Unofficial\text{-}Uber\text{-}Entry_{it}$ captures whether Uber has unofficially launched People's Uber in city *i* by month *t* (one for yes, zero otherwise) but has yet to officially launch the service. For cities that experienced unofficial Uber entry, the initial Uber entry time is the unofficial Uber entry time. The indicator $Official\text{-}Uber\text{-}Month_{it}$ equals one when Uber officially entered city *i* in month *t* (one for yes, zero otherwise). Thus, the initial Uber entry time refers to the official Uber entry for cities that did not experience unofficial Uber entry. The indicator $Post\text{-}Uber\text{-}Entry_{it}(k)$ equals one if the temporal distance between Uber's official entry into city *i* and month *t* is *k* (in three-month intervals after the official Uber entry month). As the number of posttreatment periods is limited for some of the treated cities, the estimation of $Post\text{-}Uber\text{-}Entry_{it}(2)$ and $Post\text{-}Uber\text{-}Entry_{it}(3)$ leverages posttreatment period data of a limited set of treated cities. The estimation of $Official\text{-}Uber\text{-}Month_{it}$ and $Post\text{-}Uber\text{-}Entry_{it}(1)$ is, thus, differently powered relative to $Post\text{-}Uber\text{-}Entry_{it}(2)$ and $Post\text{-}Uber\text{-}Entry_{it}(3)$. Consistent with prior work (Burtch et al. 2018), the period prior to Uber entry is omitted to avoid the dummy variable trap. Choosing different baselines yields consistent results.

In addition to city and calendar fixed effects, we include $\boldsymbol{X}_{it}$, the vector of the controls from Section 3.3.3. We include θ_i to capture monthly city-specific time trend of city *i*. The inclusion of panel-specific trends is a common approach (Besley and Burgess 2004, Wooldridge 2005) to address omitted variables biases, especially when such trends may correlate with the timing of the treatment (Angrist and Pischke 2009). As our data have a long pretreatment period, identification of city-specific trends is plausible (Wolfers 2006, Angrist and Pischke 2009). Identification of any change posttreatment, therefore, comes from deviations from preexisting trends in city *j* (Besley and Burgess 2004, Angrist and Pischke 2009). To account for potential correlation in the standard errors within city, we cluster standard errors at the city level, that is, the unit of treatment.

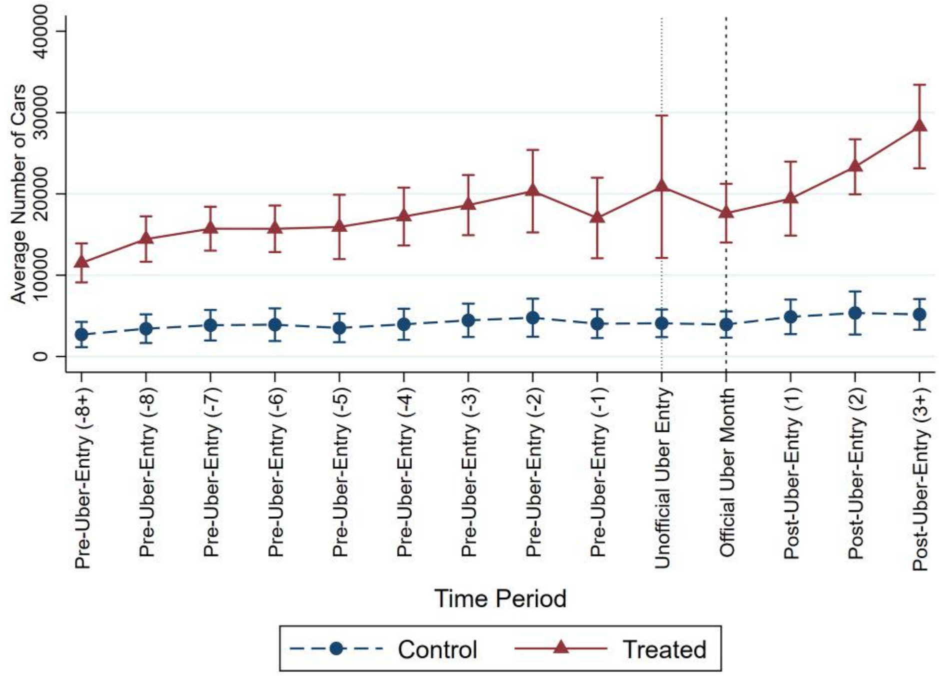
4.1. Matching

Following Bapna et al. (2018), who propose that unobserved heterogeneity may influence the likelihood of a subject being treated, the most conservative possible sample would be to exclusively use treated cities. This also assists with the inverse weighting problems described by Goodman-Bacon (2021). Yet, as there are only 14 treated cities, power issues are a material concern. We, therefore, consider counterfactual sets in which Uber does not enter. The most obvious of these is to leverage all cities in China without a purchase restriction auction. However, this poses challenges as the cities Uber enters are larger than the average Chinese city (Table 1), which undermines the similarity between treatment and control. To the extent that increased homogeneity between treatment and control increases the strength of any causal claim, we build a series of matched samples in which each treated city is matched with a city Uber does not enter.

One concern with traditional matching methods, as with propensity scoring matching (PSM) (Ho et al. 2017, Terwiesch et al. 2020), is that they do not incorporate time-varying covariates that may influence both (i) whether a unit is treated and (ii) when it is treated. To take advantage of the paneled structure of our data, we estimate a variant of a PSM, a time-dependent PSM. Results (in Online Appendix C) are consistent with a traditional PSM or a coarsened exact matching (CEM). We do so because it allows us to pair each treated city with a control city based on time-varying covariates (Lu 2005). In doing so, the propensity scores of Uber entry timing are estimated based on a discrete time logit hazard model, wherein we regress dichotomous Uber entry on the full corpus of controls from Section 3.3.3. Such a hazard model (i) allows us to also examine whether Uber entry is correlated with pretreatment changes in vehicle purchasing, thereby leading to a biased interpretation of the effect, and (ii) permits us to include both time fixed effects and time-varying controls when estimating the hazard rate (Singer and Willett 1993).

Details regarding the estimation of the discrete-time logit hazard model are in Online Appendix B. Results indicate that vehicle sales volume is not significantly correlated with Uber entry. After the discrete-time logit hazard model is estimated, hazard rates are estimated and used as propensity scores for matching. Sequential matching is performed chronologically based on the propensity score of each treated city at the time of treatment without replacement (Lu 2005). That is, starting from the earliest treatment (October 2014), for each treatment cohort (as defined by treatment month), each treated city is matched at the time of treatment with the closest untreated city based on the estimated propensity

Figure 1. (Color online) Time Trends of Number of Cars: Treated vs. Control Cities Based on Time-Dependent PSM Sample



score (i.e., hazard rate). Matched cities are then removed, and the matching process continues with the next treatment cohort.

The final matched sample includes 14 treated cities with 14 control cities. Figure 1 shows the time trends of the average number of cars in treated and control cities. For each control city, the relative time dummy used in the plot is the relative time dummy of its corresponding matching treated city to facilitate comparison. As seen, the treated and control cities share similar trends before Uber entry, but we observe a relative divergence in car sales for the treated cities compared with control cities after Uber entry. This offers further corroborative evidence for the validity of the parallel trends assumption. Raw matching statistics are in Online Table B2. It bears note that the raw matching statistics should be interpreted extremely cautiously if they are interpreted at all. The reason for this caution is that these figures are not conditioned upon the suite of covariates in the two-way fixed effects model (Equation (1)). As a result, any interpretation beyond a general increase in homogeneity between treatment and control is meaningless.

5. Results

Results are in Table 2. Recall that using only the treated cities is the most restrictive sample. We first estimate the model absent controls or city-specific trends (column (1)). We then include city-specific linear trends (column (2)). Finally, we include the controls (column (3)).

In column (1), the base model, we observe the existence of significant pretreatment indicators. This underscores the need for linear trends at the city level. In column (2), once city-level linear time trends are included, no pretreatment differences persist. As a result, all future estimations include city-level monthly linear time trends. When linear trends at the city level are included, the identification of the effect comes from whether the treatment leads to deviations from preexisting panel-specific trends (Besley and Burgess 2004, Wooldridge 2005). Note that higher order polynomial trends are insignificant and do not materially increase the variance explained. As linear trends obviate any significant pretreatment differences, high-order splines appear unnecessary. In column (2), we note a significant increase in the number of new vehicles purchased after Uber entry. This effect grows over time as is expected given Uber's reliance on network effects.

In column (3), which presents results from the complete model with both city-specific trends and controls, we again observe a lack of significant pretreatment trends and a significant increase in vehicle purchasing after platform entry, supporting Hypothesis 1b. Specifically, this effect grows over time with estimates monotonically increasing from 0.106 for *Official-Uber-Month* to 0.282 for *Post-Uber-Entry*(3+). These results provide compelling evidence of the dominance of the value-enhancement effect inasmuch as the presence of Uber significantly increases consumers' short-term propensity to purchase new vehicles. Unsurprisingly, we observe no significant effect of unofficial Uber entry on vehicle purchase. Whereas it is possible that the limited number of observations renders unofficial entry underpowered,

Table 2. Relative Time Model (OLS)

	(1)		(2)		(3)		(4)		(5)	
Dependent variable	*ln(NumCars)*		*ln(NumCars)*		*ln(NumCars)*		*ln(NumCars)*		*ln(NumCars)*	
Sample	Treated cities		Treated cities		Treated cities		TDPSM sample		TDPSM sample	
Pre-Uber-Entry(−8+)	−0.354	(0.205)	−0.0672	(0.157)	0.0304	(0.141)	0.103	(0.0793)	−0.0832	(0.0666)
Pre-Uber-Entry(−8)	−0.311*	(0.160)	−0.0601	(0.128)	0.0317	(0.109)	0.0901	(0.0818)	−0.0273	(0.0665)
Pre-Uber-Entry(−7)	−0.290*	(0.152)	−0.0633	(0.126)	0.00290	(0.118)	0.0147	(0.0866)	−0.0704	(0.0653)
Pre-Uber-Entry(−6)	−0.299*	(0.144)	−0.0997	(0.118)	−0.0557	(0.106)	−0.00520	(0.0809)	−0.0690	(0.0635)
Pre-Uber-Entry(−5)	−0.290**	(0.125)	−0.124	(0.0801)	−0.0917	(0.0759)	0.0151	(0.0742)	−0.0311	(0.0538)
Pre-Uber-Entry(−4)	−0.193*	(0.0904)	−0.0661	(0.0632)	−0.0455	(0.0591)	0.0356	(0.0699)	0.00438	(0.0526)
Pre-Uber-Entry(−3)	−0.101	(0.0927)	−0.0161	(0.0700)	−0.00161	(0.0686)	−0.000785	(0.0608)	−0.0217	(0.0542)
Pre-Uber-Entry(−2)	0.0165	(0.0655)	0.0620	(0.0549)	0.0718	(0.0547)	0.0672	(0.0460)	0.0619	(0.0450)
Baseline: *Pre-Uber-Entry*(−1) omitted										
Unofficial-Uber-Entry	0.177	(0.132)	0.0431	(0.0866)	0.0342	(0.0809)	0.116	(0.127)	−0.00757	(0.0907)
Official-Uber-Month	0.156**	(0.0704)	0.109**	(0.0444)	0.106**	(0.0456)	0.0763	(0.0540)	0.0772	(0.0530)
Post-Uber-Entry(1)	0.190**	(0.0836)	0.118**	(0.0404)	0.116**	(0.0452)	0.0341	(0.0528)	0.0424	(0.0450)
Post-Uber-Entry(2)	0.356***	(0.112)	0.221***	(0.0670)	0.214***	(0.0395)	0.139**	(0.0546)	0.129***	(0.0363)
Post-Uber-Entry(3+)	0.463***	(0.149)	0.270	(0.155)	0.282***	(0.0640)	0.186**	(0.0809)	0.186**	(0.0872)
Controls	No		No		Yes		Yes		Yes	
Observations	1,008		1,008		1,008		1,968		1,968	
R^2	0.814		0.855		0.861		0.796		0.835	
Month fixed effects	Yes		Yes		Yes		Yes		Yes	
City fixed effects	Yes		Yes		Yes		Yes		Yes	
City trend	No		Linear		Linear		No		Linear	

Notes. Robust standard errors clustered at city level are in parentheses. Controls include (i) variables related to Baidu indices, including *ln(Didi)*, *ln(Yidao)*, and *ln(Shenzhou)*; (ii) variables related to city demographics, including *ln(Population)*, *ln(Employed)*, *ln(Registered Unemployed)*, *ln(Average Wage)*, *ln(GRP Per Capita)*, *ln(Mobile Subscribers)*, *ln(Internet Subscribers)*, *ln(Highway Passenger Traffic)*, *ln(Buses)*, *ln(Bus Passenger Volume)*, *ln(Taxis)*, *ln(Road Area Per Capita)*, and *ln(Metro Length)*; (iii) *ln(Gasoline* 90 *Price)*.

***$p < 0.01$; **$p < 0.05$; *$p < 0.1$.

it is also possible that the lack of Uber's marketing did not generate the network of riders that would be necessary to incentivize vehicle purchase.

Turning to the time-dependent PSM results, we again first estimate the model absent city-specific trends (column (4)) and then include city-specific linear trends (column (5)). Results are consistent both without and with city-specific linear trends. There are no significant differences in pretreatment trends, suggesting the parallel trends assumption is not overtly violated in this sample. Further, posttreatment effects are consistently positive, indicating that short-term value enhancement dominates any decline in purchasing stemming from drivers idling their vehicles or postponing vehicle purchase. This offers further support for Hypothesis 1b. It is worth noting that the estimates of *Official-Uber-Month* and *Post-Uber-Entry*(1) become statistically insignificant, which could be because of a smaller matched sample that renders these variables underpowered.

Economically, results in Table 2 are compelling, suggesting that the introduction of People's Uber is associated with a significant increase, roughly 13%–21% across estimations, in the number of new car registrations six months after official entry. Anecdotal accounts support the size of such an estimate. Almost 100,000 drivers and 800,000 passengers registered for the service within the first month of its availability in Hefei, a city with a population of seven million (Qian 2016).

Results based on the traditional PSM and the CEM are in Online Appendix C and are consistent with the main results. The positive and significant posttreatment effects suggest that the value-enhancement effect dominates any slowdown in vehicle purchasing in the short term.

6. Robustness Checks

Next, we consider an extensive set of tests to establish the robustness of the main result. For readability, a summary of these tests is in Table 3. They include a count estimator, alternate estimations (viz., feasible generalized least square (FGLS) and panel-corrected standard errors (PCSE) estimations), a wild cluster bootstrap, alternate specifications (using monthly relative time dummies), placebo treatments, and more.

6.1. Specification Validity

6.1.1. Count Model. In our main analysis, we log the dependent variable, which allows us to interpret the effect as an elasticity. We do so because the number of cars is highly skewed. The log transformation also allows us to make the distribution closer to normal when estimating the ordinary least squares (OLS). In light of concerns raised by Silva and Tenreyro (2011), we replicate the estimations using a Poisson pseudo-maximum likelihood estimator and an untransformed dependent variable. Doing so allows us to estimate

Table 3. Summary of Robustness Tests

Concern	Test	Finding	Location
Uber may be strategically entering cities with spiking rates of vehicle sales	Examine predictors of Uber entry into a city using a discrete time logit hazard model	Lagged vehicle sales not significantly correlated Uber entry	Online Table B1
Lack of comparison between treated and untreated cities	Include matched cities without Uber entry as a control group	Results remain consistent	Table 2, Online Table C1, Online Table C2
OLS not capturing the nature of count data	Replicate analysis using count (i.e., Poisson) model	Results remain consistent	Online Table D1
Serial correlation and heteroscedasticity in standard errors	Replicate analysis using feasible generalized least square and panel-corrected standard errors estimation	Results remain consistent	Online Table D1
A small number of clusters which may result in smaller *p*-values and over-rejection	Replicate analysis using wild cluster bootstrap-based inference	Results remain consistent	Online Table D2
Incorrect operationalization of relative time dummies (using three-month intervals)	Replicate analysis using monthly relative time dummies	Results remain consistent	Online Table D3
Potential confounding factors that may be correlated with Uber entry	Run placebo test using data on prior years to check seasonality which may coincide with Uber entries	Placebo treatment (i.e., two-year lagged treatment) not significantly correlated with vehicle sales	Online Table D4
	Randomly assign Uber entry timing to treated cities	Pseudo treatment not significantly correlated with vehicle sales	Online Table D5, Online Figure D1, Online Figure D2
Competing ridesharing services entry may coincide with Uber entry	Control for search volume of keywords related to other services, excluding the months of DiDi Kuaiche entry	Results remain consistent	Table 2, Online Table D6
The positive effect of Uber entry on vehicle registrations observed may be driven by earlier treated cities and there is no effect for later treated cities	Examine the differential treatment effect of earlier versus later treated cities	The effect of the later treated cities does not differ from the treatment effect of the earlier treated cities	Online Table D7

clustered standard errors (Wooldridge 1997) and avoid the issues associated with negative binomial estimations (Allison and Christakis 2006). Results remain consistent (Online Appendix D, Online Table D1).

6.1.2. Feasible Generalized Least Square and Panel-Corrected Standard Errors. As our data have relatively long panels for a small number of cities, concerns that the standard errors may be heteroskedastic and serially correlated are warranted. Following Cameron and Trivedi (2010), we replicate the main model using FGLS estimation and PCSE, which assumes heteroskedastic errors and AR(1) autocorrelation. Results remain qualitatively similar (Online Appendix D, Online Table D1).

6.1.3. Wild Cluster Bootstrap-Based Inference. In our main analyses, we cluster robust standard errors on the city. However, when there are a small number of clusters, clustered robust standard errors may generate small *p*-values, resulting in over-rejection (Bertrand et al. 2004). A common correction for this problem is to use a wild cluster bootstrap (Cameron et al. 2008). Results of a wild cluster (Online Appendix D, Online Table D2) are consistent with prior estimates, suggesting that our inference is not affected by the number of clusters.

6.1.4. Alternate Model Specification (Month). The relative time model in Equation (1) uses three-month relative time indicators. To ensure that results are robust to the choice of time dummies, we replicate the estimation using relative time dummies at the month level. Results remain consistent (Online Appendix D, Online Table D3).

6.2. Placebo Test Using Data on Prior Years

Although we control for general seasonal trends and incorporate city-specific linear trends, it is possible that the pattern in sales organically correlates with the timing of Uber's entry for reasons unknown to researchers. Following Danaher and Smith (2014), we next execute a placebo test using data on prior years to safeguard against this possibility. In doing so, we change the "timing" of treatment to two years before treatment occurred and drop the final two years of the panel. For example, the treatment for Chongqing becomes November of 2012 instead of November of 2014. We then replicate our estimations. Results are in Online Appendix D, Online Table D4. As expected, no systemic correlation exists between pseudo-entry and vehicle sales (with the exception of a marginally significant single posttreatment indicator at $p < 0.1$). This suggests that the observed effect of Uber entry is unlikely to be driven by a structural abnormality relating to seasonality.

6.3. Random Treatment Test

We further implement two random treatment tests to examine the possibility of false significance resulting from serial correlation in the dependent variable or spurious effects (Bertrand et al. 2004). In the first, we randomize the treatment timing of each treated city. We then replicate our estimates using the pseudo treatment and store the estimates. For simplicity, we do not assign any city unofficial Uber entry. This procedure is replicated 2,000 times with different sets of random treatment. The distributions of estimated coefficients are in Online Appendix D, Online Figure D1, and the 95% intervals of the estimates are in Online Table D5. Three takeaways are apparent. First, as expected, the distributions of the pseudo effects are centered around zero. Second, *t*-tests fail to reject the null hypothesis; that is, the average estimated effects are statistically indistinguishable from zero. Third, comparing the distribution of the pseudo effects of random postentry dummies, we see that the estimates from the actual data are at the right tail of the distribution of the pseudo effects (higher than the 99% percentile), suggesting that the probabilities of obtaining similar estimates by random chance are extremely low (p-value < 0.01).

In the second, we perform a similar procedure using the time-dependent PSM sample in Section 4.1. In doing so, we randomly assign half of the cities as treated cities and then randomly assign the treatment timing within those cities. We then replicate the relative time estimates using the random treatment timing and store the estimates. This procedure is replicated 2,000 times. The distributions of coefficients estimated from the test are in Online Figure D2, and the 95% intervals of the estimates are in column (2) of Online Table D5. Results indicate the pseudo treatment is once again centered on zero, and the true estimates are unlikely to manifest by chance.

6.4. Competing Ridesharing Services—Didi Kuaiche

As discussed in Section 3.1, Uber was the largest peer-to-peer ridesharing service in China during the sample. The other major ride-hailing service, DiDi, mainly provided taxi-hailing services between 2012 and mid-2015. Further, unlike Uber, which phased its introduction, DiDi launched its discount ridesharing service nationally in May and June of 2015. Thus, we cannot exploit the same staggered entry that characterizes Uber entry. As there is no control group, any national rise in purchasing is absorbed by the time fixed effect.

Still, one implication of this mass rollout is attribution, that is, whether the surge in vehicle purchasing is a result of Uber or these other ridesharing services. Although these ridesharing competitors should be subject to the same theoretical mechanisms as Uber, to best isolate the effect of Uber alone, in addition to time fixed effects that capture universal changes across cities, we remove any period in which DiDi Kuaiche, DiDi's equivalent of People's Uber, is introduced (viz., May and June of 2015). Results are in column (1) of Online Table D6 and remain consistent. It also bears note that we separately control for the search volume of the different types of DiDi services (namely, the umbrella brand DiDi Chuxing, the ridesharing service Didi Kuaiche, and other nonridesharing services provided by DiDi). Results are in column (2), Online Table D6 and are consistent. Interestingly, we find no relationship between DiDi search volume and vehicle purchase.

6.5. Earlier vs. Later Treated Cities

Finally, it is possible that the positive effect of Uber entry on vehicle registrations observed may be driven by earlier treated cities and that there is no effect for later treated cities. To test this possibility, we create a dummy $Later_i$, which is equal to one if Uber officially entered city i after June 2015 and zero if official entry to city i occurred before or during June 2015 (the median entry point). We then interact $Later_i$ with the posttreatment dummy variables. For cities where Uber officially entered after June 2015, the relative time dummy *Post-Uber-Entry*(3+) becomes irrelevant because our data stop in December 2015. Results are in Online Appendix D, Online Table D7. Column (1) reproduces the results of column (4) of Table 2, and column (2) adds the interaction terms between $Later_i$ and the posttreatment dummies. Results indicate that the coefficients of the interaction terms (i.e., $Later_i \times$ *Post-Uber-Entry*(1) and $Later_i \times$ *Post-Uber-Entry*(2)) are statistically insignificant, suggesting that the effect of the later treated cities does not differ from the treatment effect of the earlier treated cities. Results of the primary relative time estimates remain consistent with those in column (1).

7. Empirical Extensions

Our analyses consistently show a positive and significant short-term rise in new vehicle registrations after entry. We next consider heterogeneity in the effect to uncover the underlying mechanism. In these extensions, we omit the pretreatment indicators in the interest of space. They are universally insignificant.

7.1. Effect by Brand Price Tier

A factor potential ridesharing drivers may consider when making vehicle purchasing decisions is vehicle price. As Uber requires vehicles to be valued at a minimum of 80,000 RMB, our expectation is that Uber entry only

Table 4. Relative Time Model (OLS): Effect by Brand Price Tier (Below Vs. Above 80,000 RMB)

		(1)	
Dependent variable		*ln(NumCars)*	
Sample		Treated cities	
Heterogeneous effect		Brand price tier	
Pre-Uber-entry dummies	Statistically insignificant and not reported		
	Baseline: *Pre-Uber-Entry*(−1) omitted		
Post-Uber-entry dummies	*Unofficial-Uber-Entry*	−0.260	(0.232)
(Main effects)	*Official-Uber-Month*	−0.109	(0.0829)
	Post-Uber-Entry(1)	−0.0589	(0.0795)
	Post-Uber-Entry(2)	0.00485	(0.152)
	Post-Uber-Entry(3+)	−0.0909	(0.224)
Brand tier (Baseline: below 80K RMB)	Above 80K RMB	0.640***	(0.144)
Interaction terms	*Unofficial-Uber-Entry* × *Above 80K RMB*	0.714*	(0.371)
(Moderating effects)	*Official-Uber-Month* × *Above 80K RMB*	0.446***	(0.118)
	Post-Uber-Entry(1) × *Above 80K RMB*	0.368***	(0.112)
	Post-Uber-Entry(2) × *Above 80K RMB*	0.433**	(0.177)
	Post-Uber-Entry(3+) × *Above 80K RMB*	0.608***	(0.188)
Controls		Yes	
Observations		2,016	
R^2		0.853	
Month fixed effects		Yes	
City fixed effects		Yes	
City trend		Linear	
Brand tier trend		Linear	

Notes. Robust standard errors clustered at city level are in parentheses. Controls include (i) variables related to Baidu indices, including *ln(Didi)*, *ln(Yidao)*, and *ln(Shenzhou)*; (ii) variables related to city demographics, including *ln(Population)*, *ln(Employed)*, *ln(Registered Unemployed)*, *ln(Average Wage)*, *ln(GRP Per Capita)*, *ln(Mobile Subscribers)*, *ln(Internet Subscribers)*, *ln(Highway Passenger Traffic)*, *ln(Buses)*, *ln(Bus Passenger Volume)*, *ln(Taxis)*, *ln(Road Area Per Capita)*, and *ln(Metro Length)*; (iii) *ln(Gasoline* 90 *Price)*. Key results, i.e., the interaction terms of interest, are highlighted.
***$p < 0.01$; **$p < 0.05$; *$p < 0.1$.

affects vehicles priced above 80,000 RMB with no effect on vehicles below 80,000 RMB. Whereas our data do not give us insights into either the precise car model or the price paid for vehicles, we can proxy for the cost of vehicles using the 2015 MRSP of vehicles for each automobile brand. In these estimates, we include two types of brands: (i) those whose models are all priced above 80,000 RMB ("high" price) and (ii) those whose models are all priced below 80,000 RMB ("low" price). Manufacturers who straddle this line are omitted. We then replicate the relative time model with the treated cities, adding interaction terms between the brand tier and postentry dummies. As seen in Table 4 and as expected, the effect of Uber entry is positive and significant for the high-price tiers and insignificant for the lower price tier. This supports our intuition that the increase in car purchases is primarily driven by the demand induced by Uber entry.

7.2. Effect by Vehicle Type

Thus far, results suggest that value enhancement dominates any short-term decrease in sales. Yet these results offer little insight into the mechanism behind the effect. If the value-enhancement effect is valid inasmuch as some vehicles may not be cost-effective in capturing rents, we expect the effect to be stronger for more fuel-efficient vehicles. Larger vehicles have inferior fuel economy and cost more to operate. Further, research suggests that vehicles with larger engines are more likely to be purchased by the wealthy (Wagner et al. 2009). As individuals driving on the platform are often capital constrained (Burtch et al. 2018), drivers should be more likely to purchase more fuel-efficient vehicles (Bellos et al. 2017). We then decompose our data by engine displacement: 1–1.5 (the baseline), 1.5–2, 2–2.5, and above 2.5 liters.

Results are in Table 5 and are consistent with expectation, suggesting that the effect of Uber entry is positive and significant for vehicles with engine displacement below 1.5 liters, but the effect is weaker for vehicles with engine displacement above 1.5 liters. Interestingly, we observe that the marginal effects of *Post-Uber-Entry*(1) and *Post-Uber-Entry*(2) become negative and significant for cars with engine displacement above 2.5 liters, suggesting that Uber entry might be cannibalizing the sales of vehicles with large engines although further work is needed to explore these dynamics fully. This corroborates the intuition that vehicles with larger engines may not be as cost-efficient as smaller engines when garnering rents from the platform.

7.3. Effect by Local Transportation Options

We further investigate heterogeneity in the effect based on established local transportation options. To the extent that Uber is entering a complex transportation ecosystem comprising taxis, bussing, rail, and personal automobile

Table 5. Relative Time Model (OLS): Effect by Engine Displacement

		(1)	
Dependent variable		*ln(NumCars)*	
Sample		Treated cities	
Heterogeneous effect		Engine displacement	
Pre-Uber-entry dummies	Statistically insignificant and not reported		
	Baseline: *Pre-Uber-Entry*(−1) omitted		
Post-Uber-entry dummies	*Unofficial-Uber-Entry*	0.0286	(0.0932)
(Main effects)	*Official-Uber-Month*	0.221***	(0.0441)
	Post-Uber-Entry(1)	0.296***	(0.0600)
	Post-Uber-Entry(2)	0.382***	(0.0886)
	Post-Uber-Entry(3+)	0.414***	(0.104)
Engine displacement (Baseline: 1–1.5 L)	1.5–2 L	0.509***	(0.0901)
	2–2.5 L	−0.167	(0.106)
	Above 2.5 L	−1.405***	(0.130)
Interaction terms	*Unofficial-Uber-Entry* × (1.5–2 L)	−0.000240	(0.102)
(Moderating effects)	*Unofficial-Uber-Entry* × (2–2.5 L)	0.131	(0.0947)
	Unofficial-Uber-Entry × (Above 2.5 L)	−0.179	(0.154)
	Official-Uber-Month × (1.5–2 L)	−0.164***	(0.0429)
	Official-Uber-Month × (2–2.5 L)	−0.0718	(0.0432)
	Official-Uber-Month × (Above 2.5 L)	−0.301***	(0.0606)
	Post-Uber-Entry(1) × (1.5–2 L)	−0.215***	(0.0436)
	Post-Uber-Entry(1) × (2–2.5 L)	−0.206***	(0.0418)
	Post-Uber-Entry(1) × (Above 2.5 L)	−0.424***	(0.0710)
	Post-Uber-Entry(2) × (1.5–2 L)	−0.196**	(0.0746)
	Post-Uber-Entry(2) × (2–2.5 L)	−0.214*	(0.105)
	Post-Uber-Entry(2) × (Above 2.5 L)	−0.516***	(0.138)
	Post-Uber-Entry(3+) × (1.5–2 L)	−0.178**	(0.0797)
	Post-Uber-Entry(3+) × (2–2.5 L)	−0.185	(0.130)
	Post-Uber-Entry(3+) × (Above 2.5 L)	−0.538***	(0.150)
Controls		Yes	
Observations		4,032	
R^2		0.944	
Month fixed effects		Yes	
City fixed effects		Yes	
City trend		Linear	
Engine displacement trend		Linear	

Notes. Robust standard errors clustered at city level are in parentheses. Controls include (i) variables related to Baidu indices, including *ln(Didi)*, *ln(Yidao)*, and *ln(Shenzhou)*; (ii) variables related to city demographics, including *ln(Population)*, *ln(Employed)*, *ln(Registered Unemployed)*, *ln(Average Wage)*, *ln(GRP Per Capita)*, *ln(Mobile Subscribers)*, *ln(Internet Subscribers)*, *ln(Highway Passenger Traffic)*, *ln(Buses)*, *ln(Bus Passenger Volume)*, *ln(Taxis)*, *ln(Road Area Per Capita)*, and *ln(Metro Length)*; (iii) *ln(Gasoline* 90 *Price)*. Key results, i.e., the interaction terms of interest, are highlighted.
***$p < 0.01$; **$p < 0.05$; *$p < 0.1$.

use, it is worth considering how access to these modes of transit moderates the effect. On the one hand, recent studies suggest that Uber can serve as a complement to transportation infrastructure, notably heavy rail, by solving last-mile problems (Babar and Burtch 2020). Such a relationship is intuitive as the flexible nature of ride-hailing can free consumers from the rigid end points set by rail. On the other hand, it is possible that Uber might displace taxi and bus services by providing more reliable point-to-point services and do so more efficiently with a centralized coordinating mechanism (Babar and Burtch 2020).

To examine the heterogeneous effects by transportation options, we classify cities into high versus low based on the number of buses and taxis. For city i in month t, the variable $High_{it}$ is equal to one if the number of buses or taxis of city i is above the median among treated cities in that year. We then interact this indicator with the posttreatment dummies to determine any differential effects on cities with high versus low levels of transportation options. Results are in columns (1) and (2) of Table 6. Interestingly, we find that the effect is stronger in cities with fewer transportation options. Although further work is needed to delve deeper into this phenomenon, this adds a striking nuance to the findings of Hall et al. (2018) and Babar and Burtch (2020) by underscoring the fact that Uber supplements transportation options in cities with weaker options. This corroborates recent work suggesting that transportation networks can supplement insufficient transportation networks in underserved communities (Pan and Qiu 2018).

7.4. Effect by Income

Finally, one may expect that the effect is stronger in cities where would-be drivers anticipate higher demand

Table 6. Relative Time Model (OLS): City Heterogeneity

		(1)		(2)		(3)	
Dependent variable		*ln(NumCars)*		*ln(NumCars)*		*ln(NumCars)*	
Sample		Treated cities		Treated cities		Treated cities	
Heterogeneous effect		Number of buses		Number of taxis		GRP per capita	
Pre-Uber-entry dummies	Statistically insignificant and not reported						
	Baseline: *Pre-Uber-Entry*(−1) omitted						
Post-Uber-entry dummies	*Unofficial-Uber-Entry*	0.0127	(0.0857)	0.00706	(0.0804)	0.0680	(0.0814)
(Main effects)	*Official-Uber-Month*	0.185**	(0.0686)	0.135**	(0.0579)	0.0746	(0.0445)
	Post-Uber-Entry(1)	0.166**	(0.0638)	0.198***	(0.0595)	0.0926	(0.0577))
	Post-Uber-Entry(2)	0.346***	(0.0578)	0.337***	(0.0682)	0.151**	(0.0574)
	Post-Uber-Entry(3+)	0.254***	(0.0686)	0.267***	(0.0710)	0.271***	(0.0733)
Interaction terms	*Official-Uber-Month* × High	−0.147*	(0.0750)	−0.0478	(0.102)	0.0926	(0.0788)
(Moderating effects)	*Post-Uber-Entry*(1) × High	−0.0821	(0.0745)	−0.134**	(0.0558)	0.0955*	(0.0489)
	Post-Uber-Entry(2) × High	−0.182**	(0.0611)	−0.181**	(0.0689)	0.176**	(0.0802)
	Post-Uber-Entry(3+) × High	—	—	—	—	0.173*	(0.0928)
Controls		Yes		Yes		Yes	
Observations		1,008		1,008		1,008	
Number of cities		14		14		14	
R^2		0.862		0.862		0.862	
Month fixed effects		Yes		Yes		Yes	
City fixed effects		Yes		Yes		Yes	
City trend		Linear		Linear		Linear	

Notes. High is equal to one if the number of buses, number of taxis, or GRP per capita is above median in a given year in columns (1)–(3), respectively. The main effect of High is absorbed by city fixed effects and dropped from estimation. Robust standard errors clustered at city level are in parentheses. Controls include (i) variables related to Baidu indices, including *ln(Didi)*, *ln(Yidao)*, and *ln(Shenzhou)*; (ii) variables related to city demographics, including *ln(Population)*, *ln(Employed)*, *ln(Registered Unemployed)*, *ln(Average Wage)*, *ln(GRP Per Capita)*, *ln(Mobile Subscribers)*, *ln(Internet Subscribers)*, *ln(Highway Passenger Traffic)*, *ln(Buses)*, *ln(Bus Passenger Volume)*, *ln(Taxis)*, *ln(Road Area Per Capita)*, and *ln(Metro Length)*; (iii) *ln(Gasoline* 90 *Price)*. For columns (1) and (2), *Post-Uber-Entry*(3+) × High is not estimated because of an insufficient number of observations. Key results, i.e., the interaction terms of interest, are highlighted.

***$p < 0.01$; **$p < 0.05$; *$p < 0.1$.

for ridesharing services or possess sufficient personal wealth to purchase a vehicle. To test this possibility, we explore the moderating effect of local GRP. In doing so, we split the GRP per capita at the median level. Results are in column (3) of Table 6 and suggest that there is a greater increase in vehicle purchasing in wealthier areas, relative to less wealthy areas. This is perhaps intuitive as cities with higher GRP likely possess a greater proportion of would-be drivers on the margin of purchasing a vehicle because they are more likely to both be able to afford a new car and capture the rents from the platform because of high demand for ridesharing services.

8. Discussion and Conclusion

Sharing platforms have emerged as a means of product sharing among owners with excess capacity and consumers with temporary needs. In this study, we examine the effect of ridesharing platforms on short-term changes in vehicle purchase. As such platforms imbue owners of durable goods with the possibility of extracting rents from the platform and consumers with the possibility of consuming goods at a discounted rate, the effect of platform entry on purchase is unclear. Results suggest the platform has a significant and positive short-term effect on vehicle purchases after entry. Although platform managers may argue that they are simply attracting existing owners who use the idle capacity of their vehicles to participate in the sharing economy, our findings indicate a strong value enhancement effect, whereby consumers are strategically changing stocks of held resources to capture the excess rents offered by the platform.

Not surprisingly, results indicate that the effect exclusively manifests for vehicles that meet Uber's price requirements with no material effect on vehicles below Uber's minimum price requirement. This provides validation for the proposed mechanism that the observed increase in vehicle purchase is driven by the demand induced by Uber entry. Interestingly, results indicate a stronger effect on smaller vehicles, suggesting that individuals who participate in the platform as suppliers tend to invest in vehicles that are cost-efficient. Finally, results indicate the effect is strongest in cities with relatively weaker public transit systems, which corroborates the notion that ridesharing platforms can supplement weak or insufficient systems.

This work makes several contributions. First, it contributes to the emerging literature on the sharing economy by empirically identifying how the presence of sharing platforms affects short-term durable goods purchase. Whereas theoretical work provides competing views (Jiang and Tian 2016, Benjaafar et al. 2018, Filippas et al. 2020, Abhishek et al. 2021), this paper is the first, to our knowledge, to provide empirical evidence of the effect on new private vehicle purchase. Results provide evidence of a value-enhancement effect on product ownership, suggesting that individuals on the supply side are actively changing their stock of products to participate on such platforms instead of simply exploiting at rest resources. Second, our work contributes to the stream of literature on the societal impacts of the sharing economy (e.g., Greenwood and Wattal 2017, Burtch et al. 2018) by showing that the introduction of sharing platforms creates externalities for local markets in the form of durable goods purchase and potential employment, notably in locations with slimmer transportation markets. Third, this study contributes to the stream of literature on the impacts of the sharing economy on incumbent industries by examining the impact of ridesharing on the automobile industry (Wallsten 2015, Zervas et al. 2017, Babar and Burtch 2020, Cui et al. 2020a, Chu et al. 2021).

This study has implications for both manufacturers and managers of sharing platforms. Despite manufacturers' concerns that sharing platforms might harm sales, results suggest that such apprehension is unwarranted, at least in the short term. They further suggest manufacturers may benefit from the presence of such platforms. In particular, manufacturers who offer products that are more likely to be utilized (e.g., smaller cars with better fuel economy) are likely to benefit from these platforms. Manufacturers should embrace these opportunities by manufacturing and promoting vehicles that comport with the needs of would-be drivers of these platforms. This work also yields implications for manufacturers by informing them of heterogeneity in the effect. As neither marketing dollars nor manufacturing capacity nor inventory are infinite, it is critical for firms to effectively allocate resources, that is, toward smaller vehicles and in locations with weaker transit systems, to avoid leaving money on the table. Sharing platforms might also consider collaborations with manufacturers to incentivize consumers to purchase durable goods. To the extent that such programs can be implemented through financing opportunities or shared exposure, both stand to gain.

This study also yields notable policy implications. Whereas it is inappropriate to make any overall conclusion regarding public welfare from this work, it does inform the ongoing debate regarding the legality of sharing platforms. This work directly contributes to at least three ongoing debates regarding the legality and regulation of ridesharing. The first relates to the appropriateness of the term "sharing." As mentioned, sharing economy platforms argue that market participants are simply exploiting at-rest resources in order to recoup rents from an otherwise inefficiently used asset. As a result, these firms argue that they should not be subject to the same regulatory oversight as incumbent firms (viz., taxis) because asset acquisition is not strategic. Our results indicate that this is not the case and a nontrivial number of participants are making investments in order to participate in the economy, that is, engaging in regulatory arbitrage. Whereas some innovations of the ridesharing market could easily be enacted by industry incumbents (e.g., digital solicitation), these markets are strictly regulated compared with their upstart counterparts. Policy makers should consider thoughtful changes to the regulation of taxi services to ensure fair competition.

Second, we contribute to the ongoing debate about traffic and congestion in cities. With New York becoming the first U.S. city to implement congestion pricing, significant debate and research is dedicated to whether ridesharing firms are exacerbating increasingly poor traffic conditions (Li et al. 2016). Our findings lend credence to the claim that ridesharing platforms may be contributing to these deleterious conditions by virtue of increasing the number of vehicles being purchased.

Finally, this work contributes to the ongoing debate about the nature of employment for sharing economy workers. To date, intense debates have raged regarding whether sharing-economy workers should be classified as contractors or full-time employees. This work strengthens the claim that these workers should be viewed as contractors for two reasons. First, because they are making strategic investments in their own tools, for example, the vehicle, the norm that the employer provides the entire corpus of assets needed to execute the task (sans labor and intellectual capital) is violated. Second, because the vehicle can be used across multiple platforms within the same session of work (viz., Uber and Lyft concomitantly), the norm that employees single source their employment within session is violated (again suggesting the driver is a contractor). Considering these facts in the presence of other observed benefits of the sharing economy is critical. As a result, we urge policy makers to take these findings into consideration when designing policy and regulations.

It is also worth considering two factors beyond the empirical scope of this study: generalizability and the duration of the effect. Regarding generalizability, additional work is clearly needed to determine if such value enhancement manifests in other platforms and locations. Regarding other platforms, emerging literature does suggest broad corroborative evidence. Indeed, to the extent that the presence of Airbnb in housing markets increases the rents landlords can charge (Barron et al. 2018), a value-enhancement effect for participants on the supply side of the sharing economy is emerging. Regarding other locations, it bears note that our analysis focuses on a ridesharing service introduced by Uber in China. Whereas China is a large market with 17 million new vehicle registrations in 2015, further work to understand how results generalize is critical. Some emergent work suggests that overall vehicle registrations (i.e., both new and used registrations) may fall after Uber entry in the United States (Sengupta et al. 2021). Whereas it is beyond the scope of this work to reconcile all heterogeneous findings, two possibilities exist. First, our work focuses on new vehicle sales as opposed to combining new and used vehicle registrations. Insofar as the secondary vehicle market is complex and driven by both supply- and demand-side factors, the net effect of Uber entry on used vehicle sales is also far from clear. Second, the contexts are different. As the services Uber operates are different, the Chinese and U.S. economies are growing differently to the extent, and the two markets have different levels of vehicle ownership, any number of factors might sway outcomes. Still, these differences are speculative. To the degree that anecdotal evidence of similar value-enhancement effects in other economies exists, for example, India (Aggarwal 2017), future work to understand the nuanced mechanics of these markets is critical.

Our study is subject to other limitations that offer rich opportunities for future work. With regard to the duration of the effect, because of the sample window, we cannot observe purchasing in the long term. However, given that Uber China was acquired by DiDi in August of 2016, there is a limited time window that can be exploited. This is a limitation that offers fruitful avenues for future research. Theoretically, it is possible that the effect persists in the long term if Uber drivers accelerate the utilization of their durable goods, but nondriver consumption and churn of vehicles remain constant. This is particularly striking if ridesharing persists in supplementing public transit (Babar and Burtch 2020), notably given historic resistance to investing in public infrastructure. Alternately, if the substitution is only between drivers and nondrivers, it is plausible that the spike in sales reflects only drivers tooling up to join the platform and subsequently re-equilibrate when purchase patterns stabilize.

Next, it is worth noting that, whereas People's Uber was the dominant ridesharing service during our sample, DiDi introduced a competing service, DiDi Kuaiche, near the end of our sample. Whereas DiDi is subject to the same theoretical mechanisms as People's Uber, we do attempt to control for the potential effect of the introduction of DiDi Kuaiche on automobile purchase by using search volume data to proxy for the popularity of DiDi. Still, there is ambiguity regarding the degree to which other platforms may contribute to the effect. Third, we are unable to provide a robust analysis of the effects of platform entry on vehicle consumption and have little insight into how the temporary incentives Uber uses to seed its network might influence vehicle purchase. These are data limitations. To the degree that increased consumption depreciates vehicles more quickly, thereby stimulating additional purchase, consumption is likely of little concern. Yet, inasmuch as temporary incentives to drive for the platform (e.g., guaranteed fares, etc.) may artificially stimulate vehicle adoption in the short rather than long term, additional work is clearly needed to understand how these incentives work. Finally, it is important to note that this study addresses only part of the overall effect of sharing platforms. As stated, it is improper to draw any conclusions from this research about the totality of the public welfare loss or gain that comes from these platforms. Instead, we hope that the sharing economy continues to evolve as an area of work, and future work can develop a comprehensive understanding of both the demand and supply sides of the sharing economy.

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